

Technical Specification of LT Shunt Capacitor

1.0 Scope

This specification describes manufacturing, testing, insurance transportation, supply, installation and commissioning of three phase delta connected 433V, 50HZ outdoor type, self healing, explosion proof metalized polypropylene, filled with dry inert gas / Nitrogen gas LT fixed shunt capacitors, intended for power factor improvement and with all accessories & allied components (like clamps & connectors, earth flat, connecting cable etc.) to be installed on LT side of DTR & above the distribution box of various ratings by way of suitable clamp with pole.

2.0 Service Conditions :

a) Maximum Ambient Temperature (° C)	: +55°C
b) Minimum Ambient Temperature (° C)	: - 4°C
c) Max. & Min Relative Humidity (%)	: 100 & 50
d) Maximum annual rain fall (mm)	: 2000
e) Maximum Wind Pressure (Kg/m ²)	: 150
f) Maximum wind velocity (m/sec)	: 50
g) Isoceraunic Level (days/year)	: 200
h) Maximum altitude above mean sea level (Meter)	: 1000
i) Seismic Level (Horizontal acceleration)	: 0.3 g

3.0 Operating conditions :

a) Nominal System Voltage	: 433 V ± 10%
b) Normal frequency	: 50 Hz
c) No. of phases	: 3
d) System operating hours	: Continuous
e) Harmonic contents	: Harmonic generating loads should be less than 15% of the total load at the trail load intended for PF correction

4.0 Applicable Standards :

Unless otherwise stipulated in the specification, capacitors shall be comply with the latest version of IS: 13340 : 2012 for self healing type, IS 13341 : 2012 for requirement for aging test, self healing test and destruction test.

5.0 Basic Requirements:

5.1 The capacitor unit shall have multilayer metalized polypropylene film (MPP).

5.2 For easy identification of damaged units with naked eye; Visual Fault Indicator shall be provided.

5.3 The capacitors are to be Fixed type, fuse less & self Healing type capacitor..

5.4 Power loss: The power loss in capacitors shall not exceed 0.2 Watt/kvar

5.5 Discharge Device:

- i. Suitable discharge device shall be connected across the capacitor units in accordance with the provision of IS : 13340. The discharge device shall reduce the residual voltage from the rated voltage to 75 volt or less within 3 min after the capacitor is disconnected from the source of supply.
- ii. Capacitor shall comply with the relevant general safety regulations for power installation as per Indian Electricity Rules 1956.
- iii. The protective cap and terminal block should entirely eliminate hazards increase of accident contact.

5.5 Earth Connection: At the base of capacitor units shall be provided with suitable earthing arrangement by GI wire in line with normal earthing practice preferably in form of proper size stud bolt having 16 mm outer (projected) length with toothed washer hexagonal nut & two nos. ring washer shall be provided in such a way that it should also serve facility for clamping capacitor unit on LT poles. The outgoing terminals of capacitor unit should be covered by a UV resistance cap. The capacitor & unit with cap should be conform to IP: 54 degree of protection. Earthing is totally bidders scope.

5.6 Protective device:

There shall be in built protection device(s) / pressure sensitive in the event of any fault/abnormal condition inside the capacitor unit. Occurrence of such condition must be indicated by any mechanism / symbol outside the capacitor unit and shall be visible from the ground to facilitate identification and replacement the faulty unit. Details of such mechanism should be furnished along with Bid & Reference Standard is to be mentioned.

6.0 General requirements :

LT Shunt Capacitor

1.0 Permissible over load: The maximum permissible overloads with regard to voltage, current and reactive Output shall conform to IS 13340 : 2012

1.1 CURRENT:

Shall be suitable for continuous operation at r.m.s. Line current 1.30 times the current occurs at rated sinusoidal voltage and rated frequency excluding transients (as per IS 13340 for self healing capacitor).

1.2 VOLTAGE:

The permissible overloads shall not exceed limits set by IS : 13340 for self healing type capacitors. Capacitor shall be suitable for prolonged operation at r.m.s. voltage between terminals not exceeding 1.1 times the rated voltage.

3.0 INRUSH CURRENT

Should be 200 times I nominal current.

4.0 BASIC IMPULSE LEVEL (BIL):

The rated basic impulse level of switch to earth as also across the open terminals shall be 03KV.

5.0 Dielectric Loss - $\leq 0.2W / KVAR$

6.0 Total capacitor Loss - $\leq 0.6W / KVAR$

7.0 Mean Life (expected) - Not less than 1,00,000 hrs. The failure rate should be within 3% (indicating the capacitors failed due to drift in accuracy) however the bidder shall replace such failed units free of cost.

8.0 INSTALLATION & POWER SUPPLY:

- LT shunt capacitor shall be installed on DTR structure through a suitable clamping arrangement. Clamp should have enough strength to hold the capacitor for long period. A suitable size of ISI marked with UV protection wire reputed marks, per phase should be used for connecting the capacitors with phase wise busbar arrangement in LT distribution kiosk.
 - DTR structure are PCC poles with rectangular cross section or steel section like RS Joist, M.S. Rails.
 - Ten (10%) percent compensation of DTR capacity may be allowed for determination of capacity of Capacitor.
 - a. 100 KVA DTR : 10 KVAR
 - b. 63 KVA DTR : 7.5 KVAR &
 - c. 25 KVA DTR : 3 KVAR
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9.0 DESIGN AND CONSTRUCTION REQUIREMENTS:

1. The capacitor shall be three-phase construction as per standard design of the manufacturer with DELTA(Δ) configuration.
2. The capacitor shall be suitable for outdoor installation and shall have sealed weather proof type construction.
3. The total capacitor unit shall be provided with a mechanical indicator to show the contact position in open/closed position.
4. All nuts, bolts and mounting structure shall be hot dip galvanized.
5. Terminal Cable size & length - Four meter multi stranded 4 sqmm CU wire. Housing shall be complied with IP 54

10.0 CONTROL PARAMETER:

The bidder shall maintain all control parameter as per requirement of WBS EDC.

11.0 MARKING

The capacitor shall be provided with a name plate legibly and indelibly marked with the following:

- a) Name of the manufacturer
- b) Type, designation and serial number
- c) Rated voltage and current
- d) Rated frequency
- e) Number of poles
- f) Rated Voltage
- g) Rated current

12.0. TESTS:

The switch shall be subjected to the following tests in accordance with the IS: 13340 : 2012

12.1 TYPE TESTS:

- a) Thermal stability test.
 - b) Measurement of the tangent of the loss angle ($\tan \delta$) of the capacitor at elevated temperature.
 - c) Voltage test between terminals
 - d) Voltage test between terminals and container.
 - e) Lightning impulse voltage test between terminals and container.
 - f) Discharge test.
 - g) Ageing test.
 - h) Self-healing test.
 - i) Destruction test.
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12.2 ROUTINE TESTS:

- a) Capacitance measurement and output calculation.
- b) Measurement of the tangent of the loss angle ($\tan \delta$) of the capacitor.
- c) Voltage test between terminals.
- d) Voltage test between terminals and container.
- e) Test of the internal discharge device.
- f) Sealing test.

12.3 ACCEPTANCE TESTS:

- I. Visual examination
- II. Test for Capacitance and Output.
- III. Capacitor Loss Tangent measurement.
- IV. Insulation resistance
- V. AC Voltage test between terminals.
- VI. AC Voltage test between terminals & container.
- VII. Test for Discharge device.
- VIII. Sealing Test.

12.4 The equipment offered by manufacturers shall comply with general safety regulations.

12.5. TYPE TEST REPORT:

The bidders shall furnish detailed type test reports of the offered LT shunt Capacitor for the tests as per relevant IS mentioned in this specification. All these Type Tests shall be carried out at laboratories that are accredited by the National Accreditation Board of Testing and Calibration Laboratories (NABL) of Government of India. These tests should have been carried out within 5 years prior to the date of opening of this tender.

13.0 DOCUMENTATION:

Bidders shall furnish following drawings & documents along with bid.

13.1 Dimensional drawing of item offered indicating all fittings.

13.2 Relevant catalogues

13.3 GTP as per Schedule-I,

3134 Bill of material as per Schedule-II



SCHEDULE: I

GUARANTEED TECHNICAL PARTICULARS FOR LT SHUNT CAPACITOR

(To be furnished by the Manufacturer)

Sr. No.	Particulars
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1.	Name of the Manufacturer :
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2.	Type of capacitor :
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3	Indian Standard according to which the capacitor cells are manufactured & tested. (IS-13340 : 2012) :
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4.	Name of the insulating materials used in the capacitor with class / grade etc. :
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5.	Service conditions maintained or not (Yes / No) :
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6.	Operating conditions maintained or not (Yes / No) :
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7.	Nominal System Voltage in V :
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8.	Rated Voltage of capacitor cells in V :
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9.	Reduced Discharge voltage with time :
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10.	General Safety complied or not :
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11.	Protective Cap & Terminal block eliminates Hazards or not :
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12.	Permissive Overload in Current :
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13.	Permissive Overload in Voltage :
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14.	Inrush Current in terms of I nominal current :
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15.	Basic Insulation level (BIL) in KV :
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16.	Dielectric Loss (W/ KVAR) :
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17.	Mean Life (expected) in Hrs. :
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18.	Capacitor configuration :
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19.	Housing shall comply with IP 54 (Yes / No) :
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20.	Rated capacity of three phase capacitor :
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8.	Rated Frequency of capacitor cells (50Hz)
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9. Power loss in capacitor (in Watts)
10. Whether suitable internal discharged device provided or not
11. Whether suitable external earthing terminal provided or not
12. Thickness of the CRCA container sheet of capacitor cell
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13. Size of the capacitor cell
14. Net weight of capacitor cell
15. Whether Type test reports of the specification are submitted
16. Any other particulars which the bidder may like to give

SCHEDULE: II

BILL OF MATERIALS

(TO BE FURNISHED BY BIDDER)

The following equipments /materials with suitable ratings listed below but not limited to, as per approval of WBSEDCL shall be supplied per DTR:

Sl. No.	Descriptions	Quantity
1.	Capacitor Unit, 3ph (including all accessories)	1set

